



清华大学电子工程系
Department of Electronic Engineering, Tsinghua University

严谨 · 勤奋 · 求实 · 创新

数字系统设计

01-课程介绍

刘勇攀 & 贾弘洋

2025/02/18



课程教师



清华大学电子工程系
Department of Electronic Engineering, Tsinghua University

任课教师：贾弘洋 & 刘勇攀

邮箱：hjia@tsinghua.edu.cn; ypliu@tsinghua.edu.cn

电话：96637

办公室：罗姆楼7层710；罗姆楼4层4-302



课程助教：汤宸 & 疏炫深 & 王与进

邮箱：tangc16@126.com; {shuxs23, yujin-wa24}@mails.tsinghua.edu.cn

电话：; ;

办公室：罗姆楼7层7-303; 罗姆楼7层708





1) 课程定位

- a) 硬件与软件的桥梁
- b) 打造适合深度学习的硬件

2) 课程内容

- a) 课程讲座、实验Lab和口头报告
- b) 论文阅读报告
- c) 工业界讲座

人工智能的应用需求



Cloud AI

大数据集/训练

大模型

快速响应

高性能

$>>10\text{kW}$

$\sim 1\text{kW}$

$\sim 100\text{W}$

$\sim 10\text{W}$

$<1\text{W}$

千亿设备规模^{*1}

- 低功耗
- 低延时
- 高能效

Edge AI

^{*1} : from Tractica.



人工智能三要素



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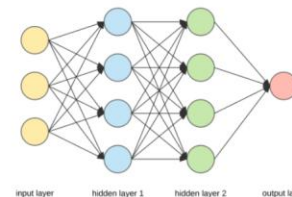
**Bigger
Data**



**More
Compute**



**Better
Algorithm**



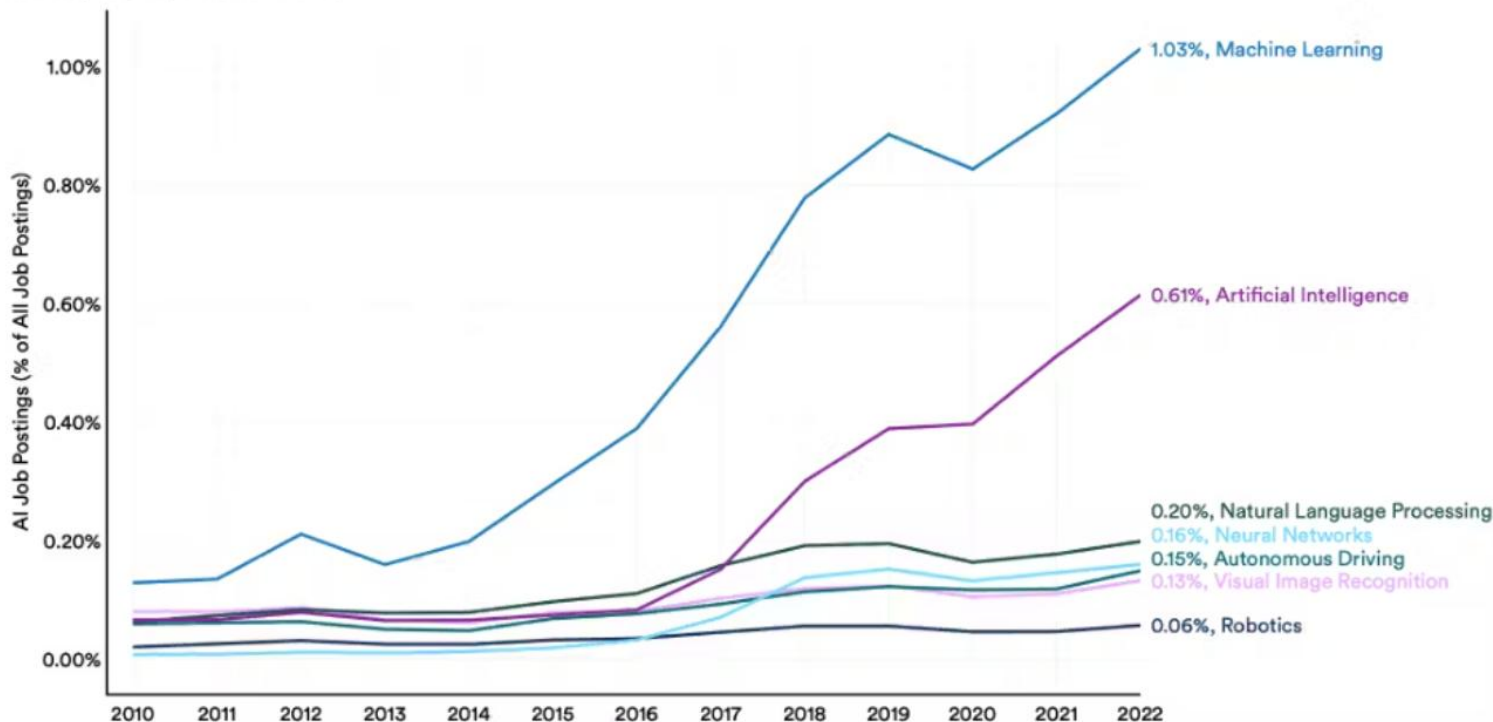
人工智能工作需求趋势



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AI Job Postings (% of All Job Postings) in the United States by Skill Cluster, 2010–22

Source: Lightcast, 2022 | Chart: 2023 AI Index Report

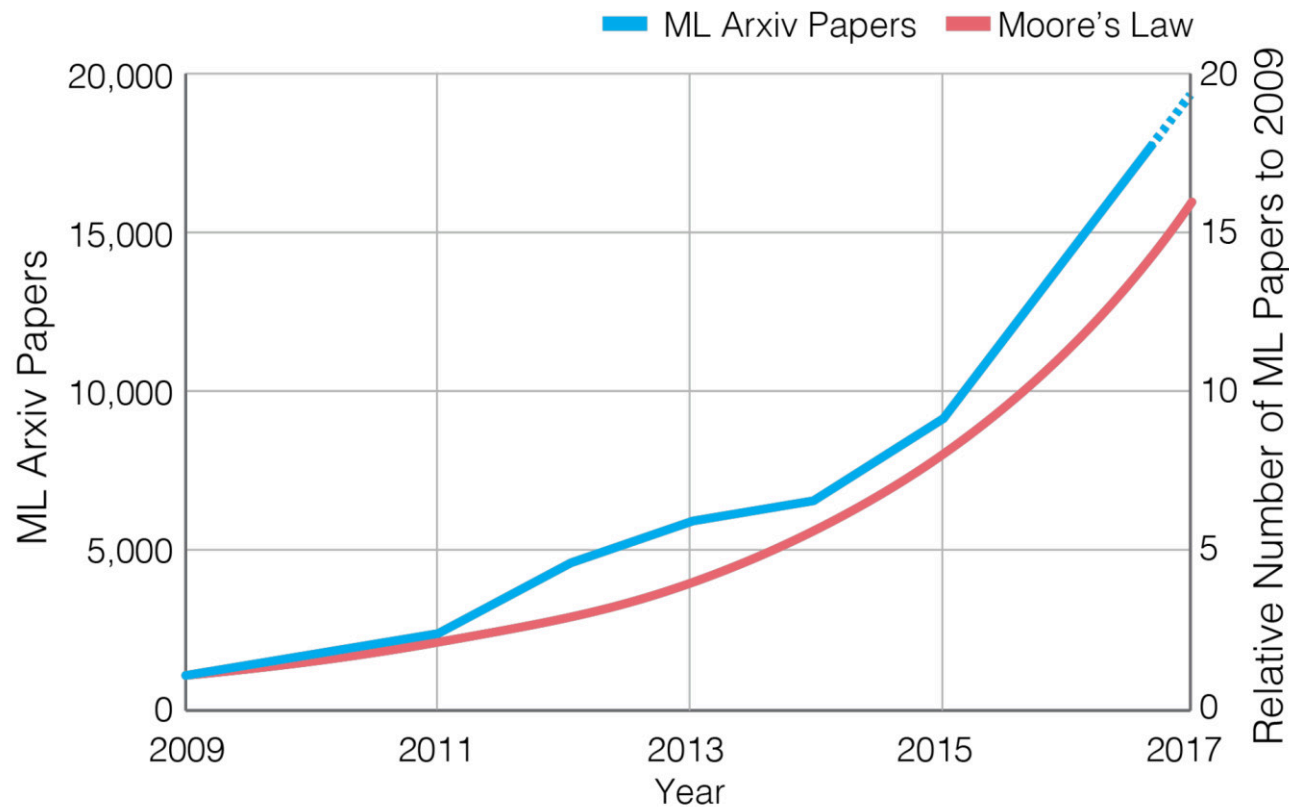


Source: Lightcast

人工智能论文发表趋势

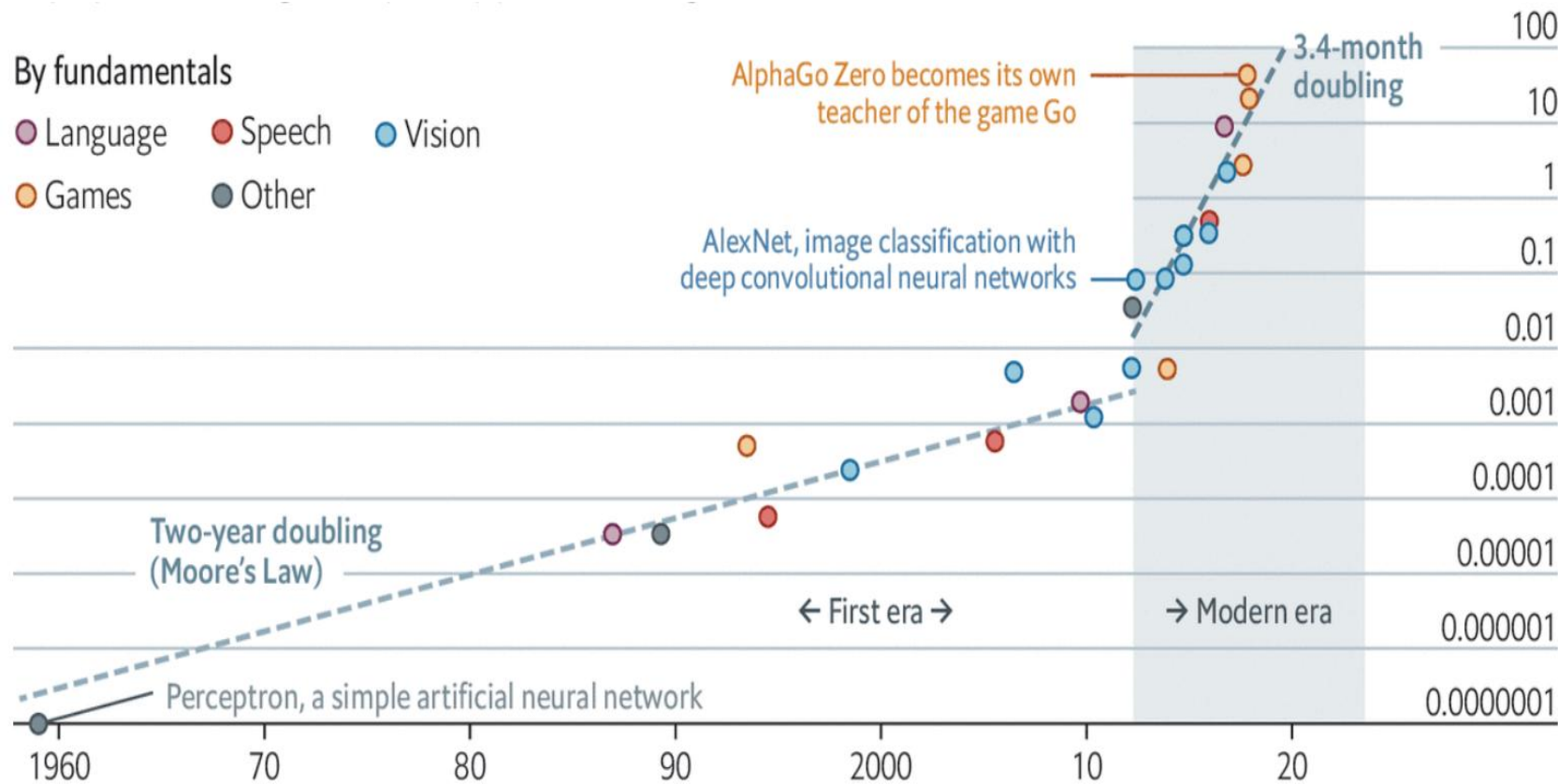


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From: J. Dean, 2018

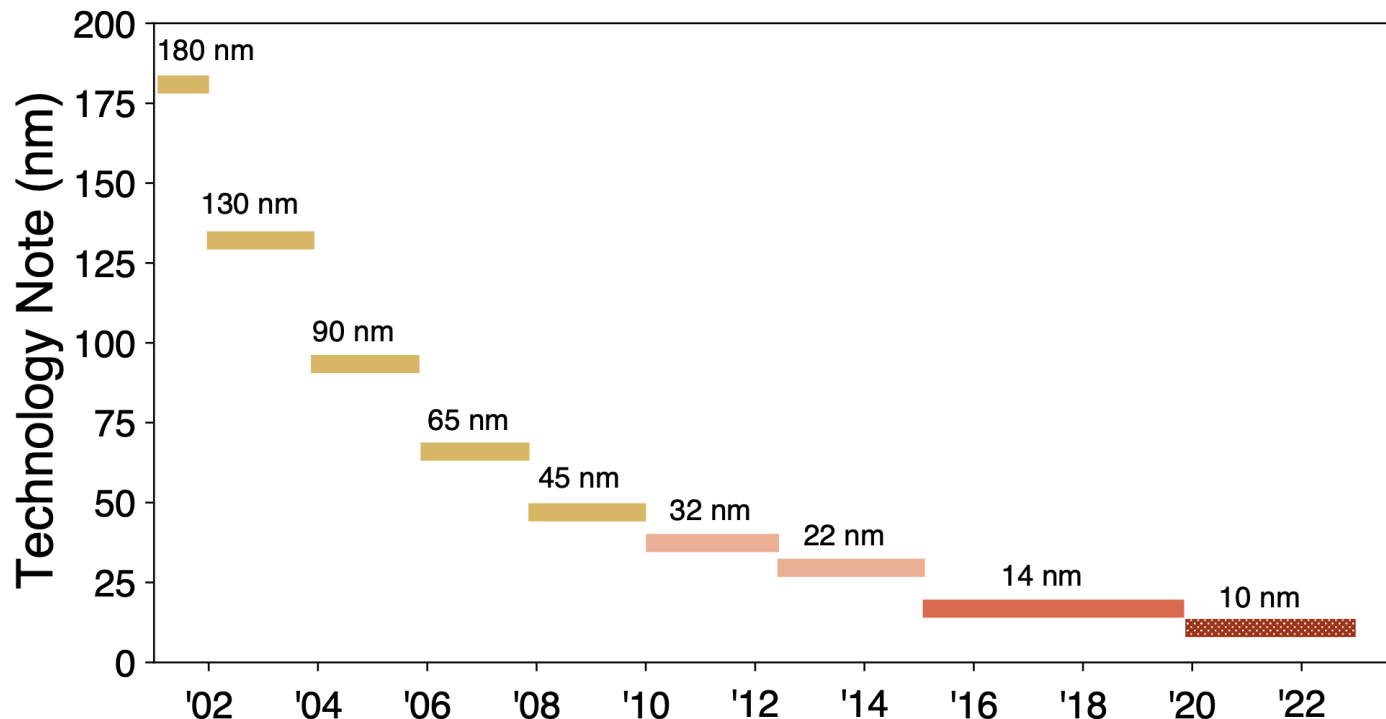
人工智能的算力需求



Source: OpenAI

*1 petaflop = 10^{15} calculations

摩尔定律放缓



低功耗智能设备的需求

Autonomous Driving

6 full HD cameras @30fps

10W budget

Model: EfficientNetV2/frame (under est.!)

➔ 2TOPs/frame, **~500 TOPs**

➔ 50TOPs/W



Augmented Reality Glasses

Stereo HD + eye tracking camera @30fps

100mW budget

Model: EfficientNetV2/frame (under est.!)

➔ 1TOPs/frame, ~50 TOPs

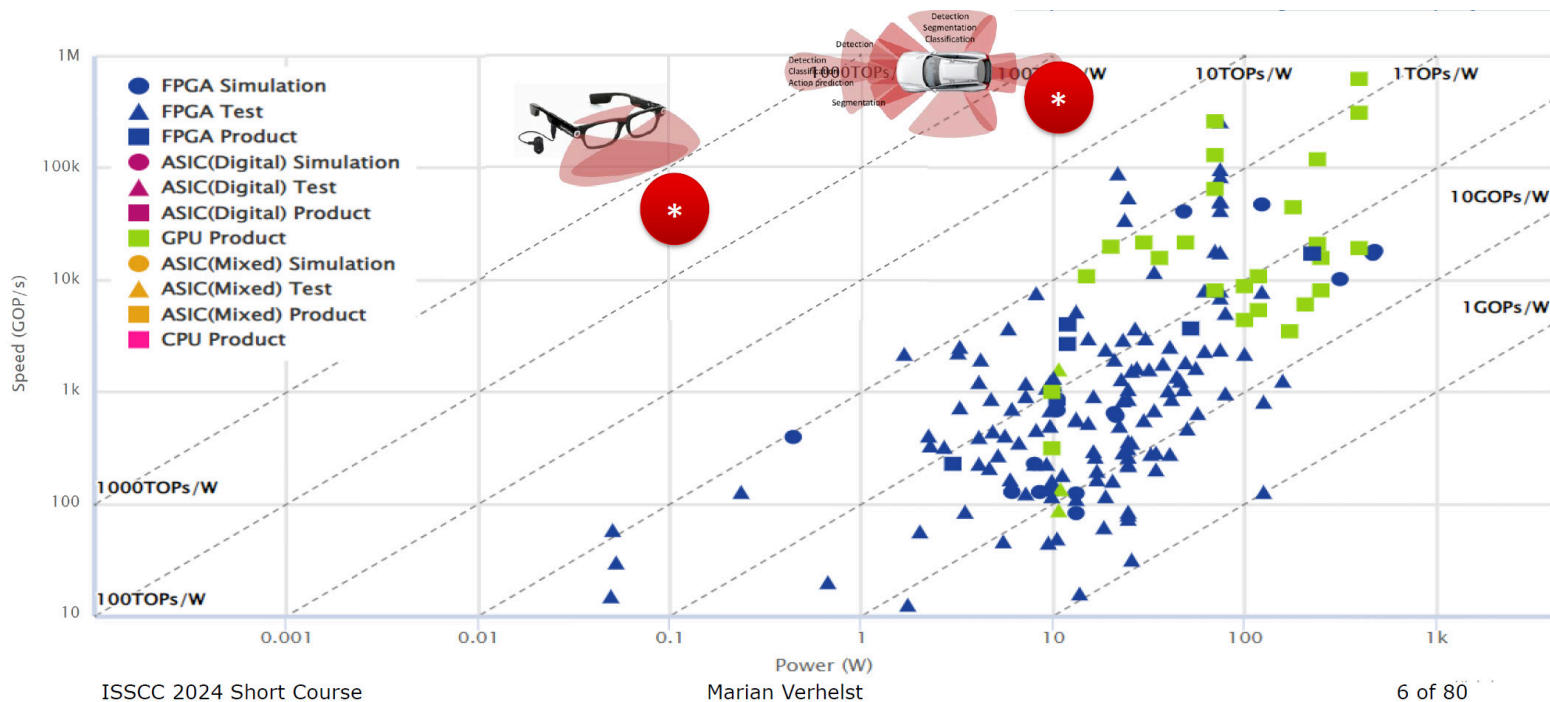
➔ **500TOPs/W**



Marian Verhelst

5 of 80

通用解决方案的问题



ISSCC 2024 Short Course

Marian Verhelst

6 of 80

© 2024 IEEE
International Solid-State Circuits Conference

<https://nicsefc.ee.tsinghua.edu.cn/project.html>

通用解决方案与需求存在2-3个数量级差距！

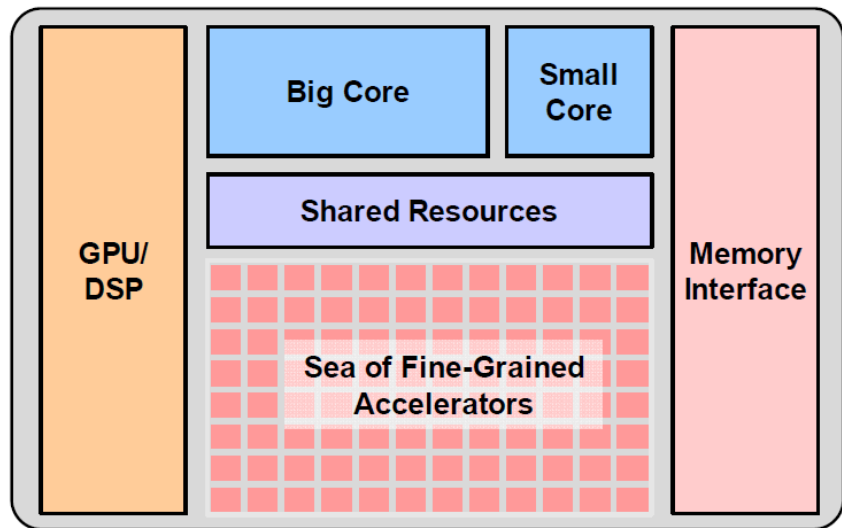


神经网络专用处理器

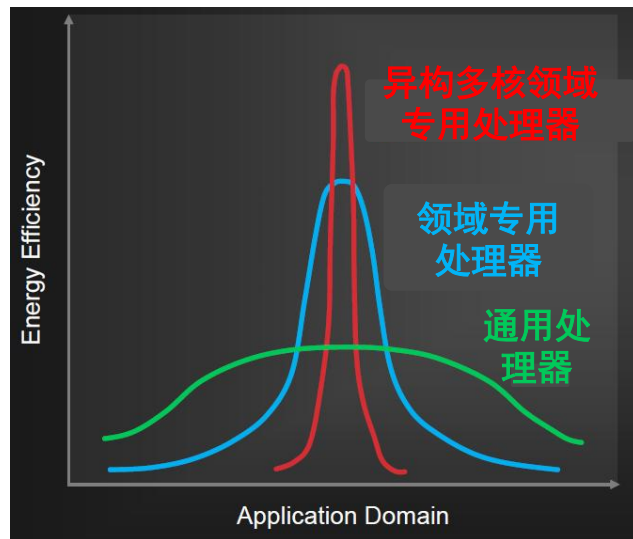
近十年算力系统芯片 (SoC) 的发展趋势



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TSMC: 加速器海



专用处理器实现高能效

发展趋势: CPU + GPU + 异构多核领域专用处理器

领域计算加速器的复兴

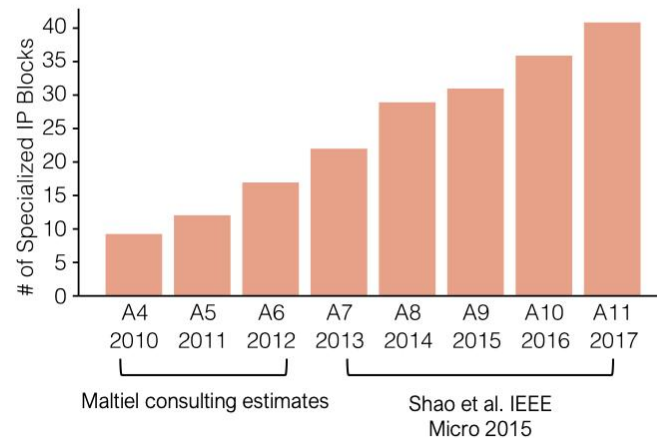
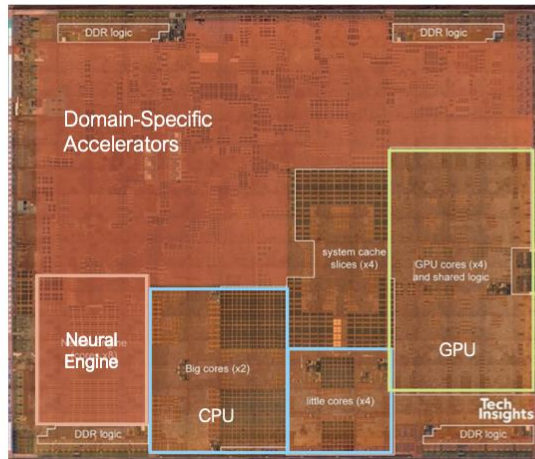


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- Customized hardware designed for a domain of applications.



iPhone XS,
2018





Conclusion: A New Golden Age

- End of Dennard Scaling and Moore's Law
⇒ architecture innovation to improve performance/cost/energy
- Security ⇒ architecture innovation too
- Domain Specific Languages ⇒ Domain Specific Architectures
- Free, open architectures and open source implementations
⇒ everyone can innovate and contribute
- Cloud FPGAs ⇒ all can design and deploy custom "HW"
- Agile HW development ⇒ all can afford to make (small) chips
- Like 1980s, great time for architects in academia & in industry!

智能芯片的兴起



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The New York Times

Big Bets on A.I. Open a New Frontier for Chip Start-Ups, Too

By Cade Metz

Jan. 14, 2018

Today, at least 45 start-ups are working on chips that can handle speech and self-driving cars, and at least five of them have raised more than \$100 million from investors. Venture capitalists invested more than \$1.5 billion in chip start-ups last year, nearly doubling the amount invested two years ago, according to the research firm CB

JII INSIGHTS

2017年-2021（1-4月）年我国人工智能芯片行业投融资情况



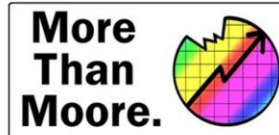
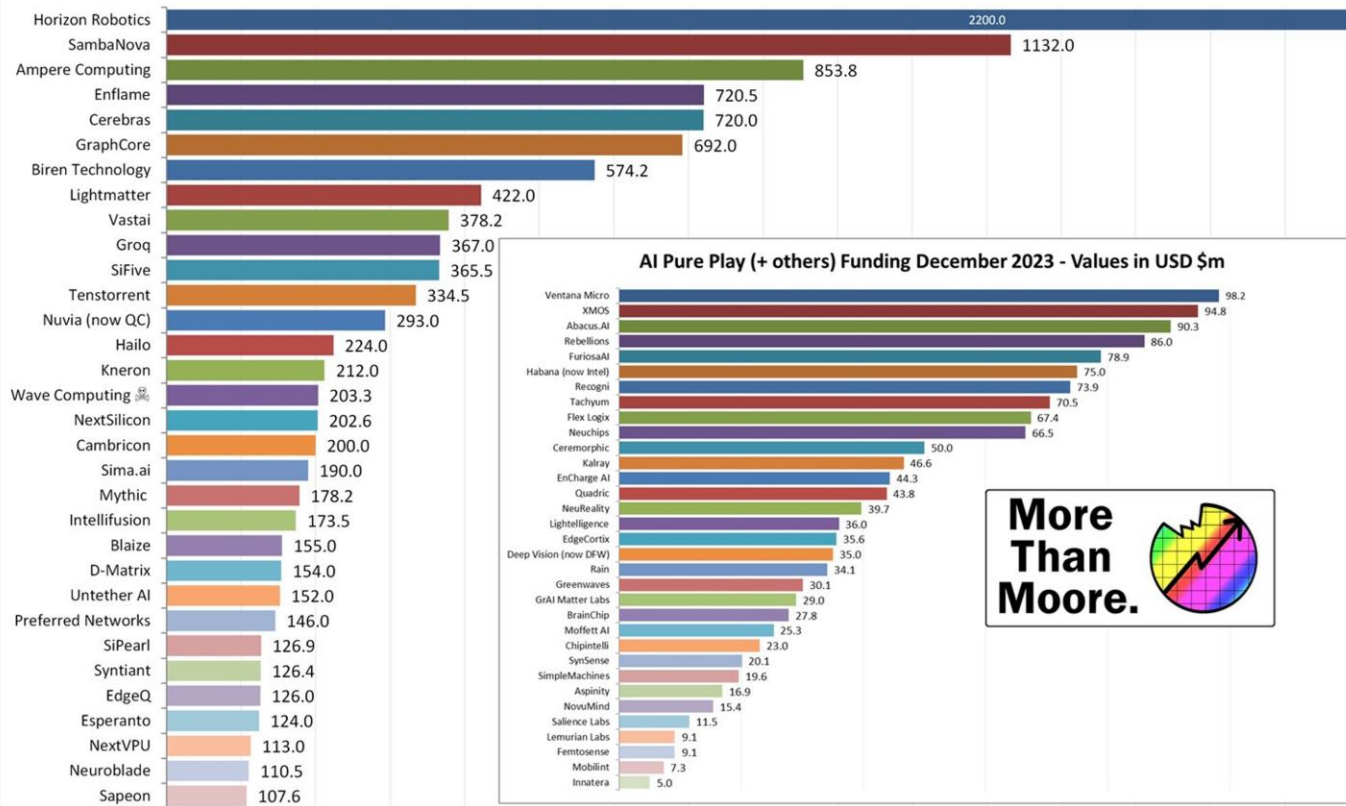
资料来源：综合IT桔子、集微网

智能芯片的兴起



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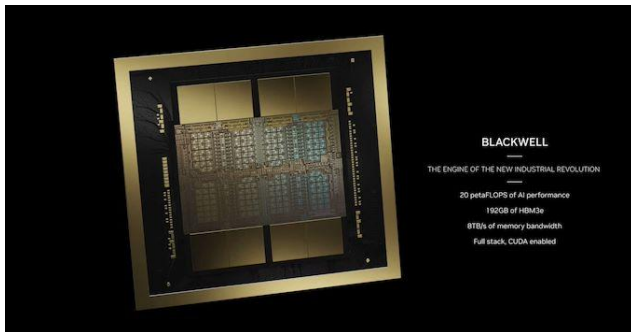
AI Pure Play (+ others) Funding December 2023 - Values in USD \$m



/ NVIDIA GPU B200 vs H100

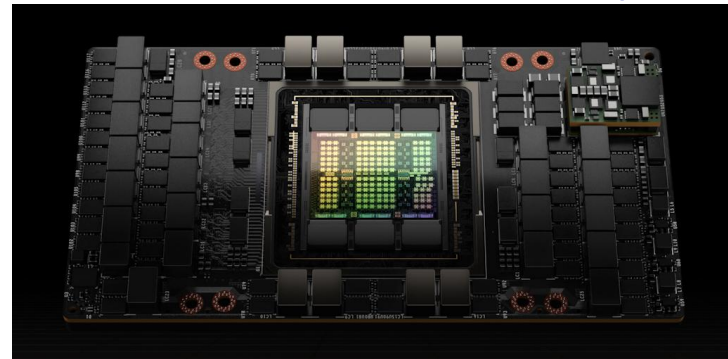
- >1500 mm²
- 208 billion transistors
- TSMC's N4 process
- 1000 Watt TDP
- 10000 TFLOPs FP8

performance **B200 (2024)**



- 814 mm²
- 80 billion transistors
- TSMC's N4 process
- 700-800 Watt TDP
- 3958 TFLOPs FP8

performance **H100 (2022)**



NVIDIA GPU 的算力提升

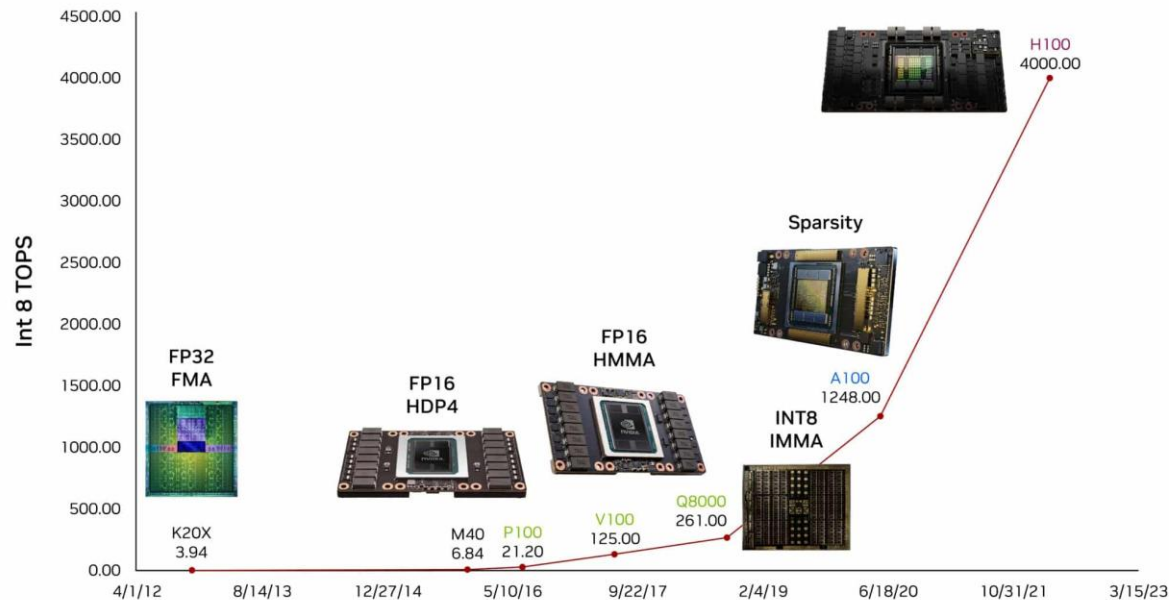


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Gains from

- Number Representation
 - FP32, FP16, INT8
 - (TF32, BF16)
 - ~16x
- Complex Instructions
 - DP4, HMMA, IMMA
 - ~12.5x
- Process
 - 28nm, 16nm, 7nm, 5nm
 - ~2.5x
- Sparsity
 - ~2x
- Model efficiency has also improved – overall gain > 1000x

Single-Chip Inference Performance - 1000X in 10 Years



[Bill Dally, Trends in Deep Learning Hardware]

NVIDIA GPU 的算力提升



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数字系统设计

现代体系结构

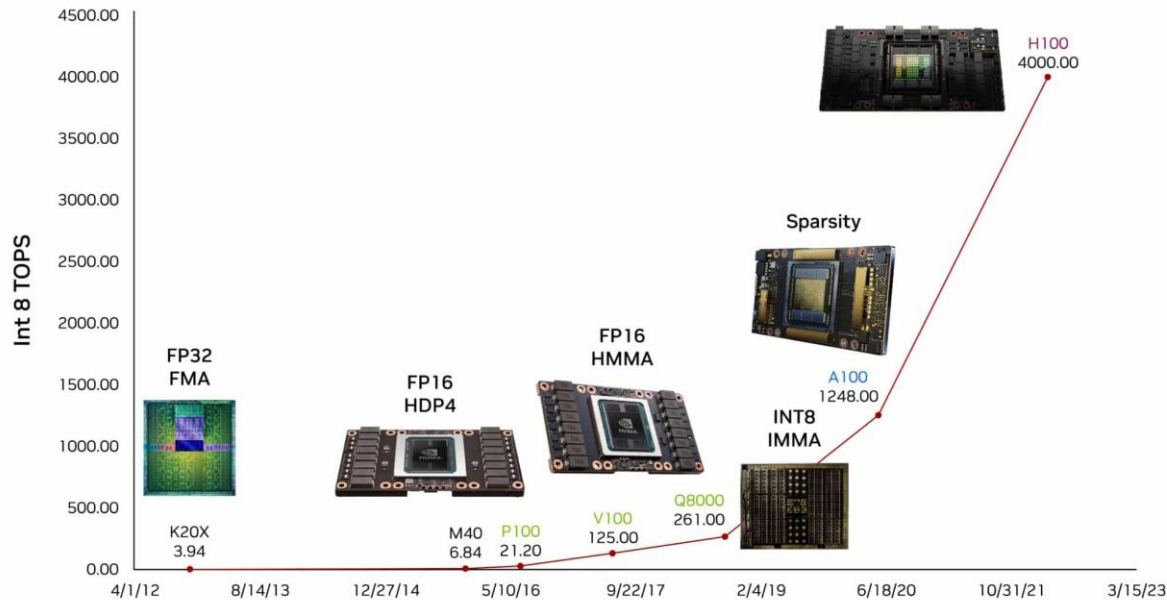
数字系统设计

半导体器件

数字集成电路

数字系统设计

Single-Chip Inference Performance - 1000X in 10 Years



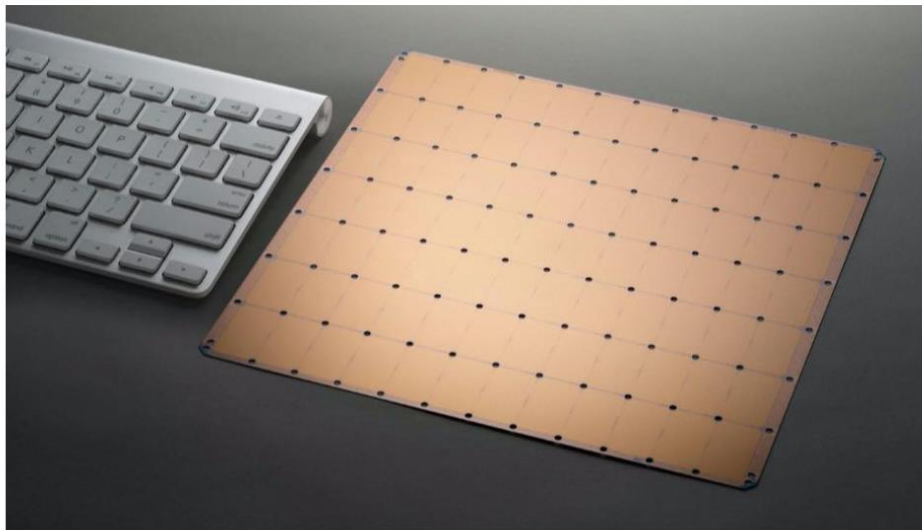
[Bill Dally, Trends in Deep Learning Hardware]

晶圆级智能芯片 Cerebras



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- Largest Chip Ever Built!
- 46,225 mm² silicon
- 1.2 trillion transistors
- 400,000 optimized AI cores
- 18 GB of on-chip memory
- TSMC 16nm process

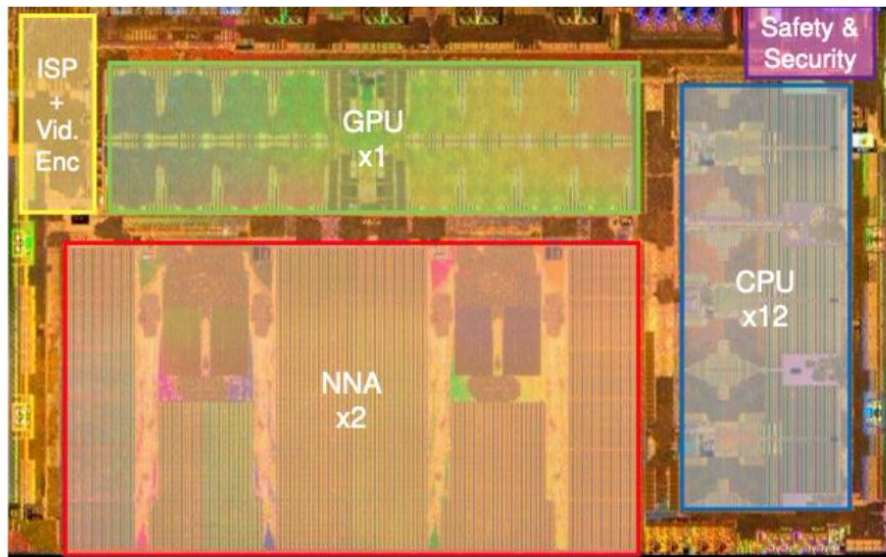


/Tesla 自动驾驶芯片



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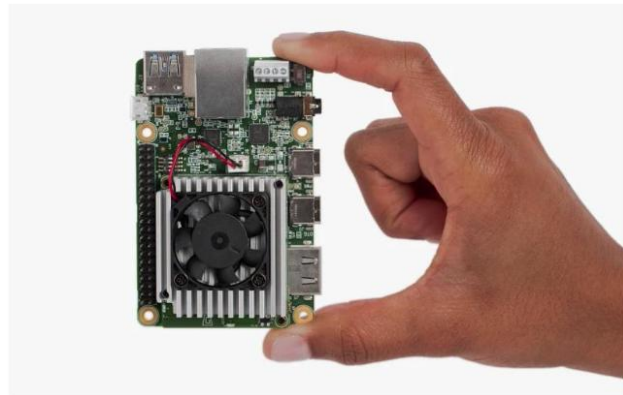
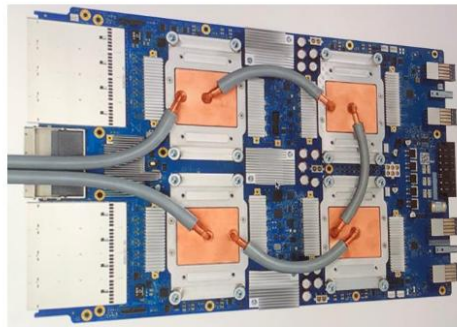
- 2 independent instances
- 2Ghz+ Design
- 32 MB SRAM / instance
- 96*96 MACs



谷歌TPU



- Systolic-array-based architecture
 - V1: Inference only
 - V2: Training with bfloat
 - V3: 2X powerful than v2
 - V4: Improve scalability by optical circuit switches
- Edge TPU
 - Coral Dev Board
 - 4 TOPS
 - 2 TOPS/Watt
 - Supports TensorFlow Lite

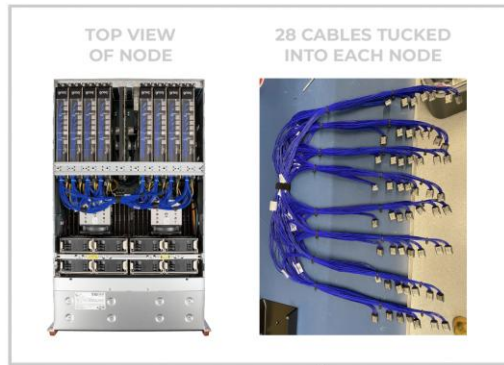
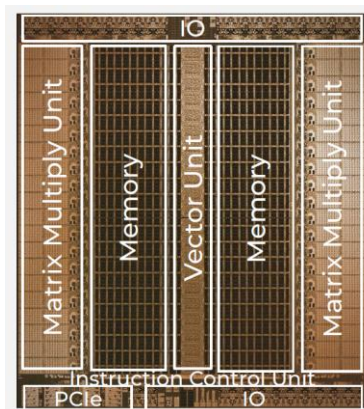


/Groq大模型处理单元 (LPU)



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- VLIW架构, 144条指令并行
- 单片750 TOPS (INT8) @ 900MHz 16nm
- 完全依靠片上SRAM进行大模型存储
 - 单片220 MB片上SRAM存储, 55 TBps 片上带宽
- 通过多芯片互联提升算力/存储
- 深度的模型-软件-硬件协同
 - Llama 3.3 70B
 - 276 Token/s



GroqChip™

GroqCard™

GroqNode™

GroqRack™

/ 测试基准和应用标准



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Fair and useful benchmarks for measuring training and inference
performance of ML hardware, software, and services.

<https://mlperf.org/>



Machine learning is no different. ML and artificial intelligence have been around for decades, but even today the technology is fragmented, bespoke, and poorly understood. We believe that we can unlock the next stage of AI/ML adoption by creating useful measures of quality and performance, large scale open data sets, and common development practices and resources. These three pillars will help to democratize machine learning and enable its widespread incorporation into new products and services in fields such as health, safety, and communication that deliver benefits to everyone.

/ 训练测试基准



Area	Benchmark	Dataset	Quality Target	Reference Implementation Model
Vision	Image classification	ImageNet	75.90% classification	ResNet-50 v1.5
Vision	Image segmentation (medical)	KiTS19	0.908 Mean DICE score	3D U-Net
Vision	Object detection (light weight)	COCO	23.0% mAP	SSD
Vision	Object detection (heavy weight)	COCO	0.377 Box min AP and 0.339 Mask min AP	Mask R-CNN
Language	Speech recognition	LibriSpeech	0.058 Word Error Rate	RNN-T
Language	NLP	Wikipedia 2020/01/01	0.72 Mask-LM accuracy	BERT-large
Commerce	Recommendation	1TB Click Logs	0.8025 AUC	DLRM
Research	Reinforcement learning	Go	50% win rate vs. checkpoint	Mini Go (based on Alpha Go paper)

Each benchmark measures the wallclock time required to train a model on the specified dataset to achieve the specified quality target

/推理测试基准



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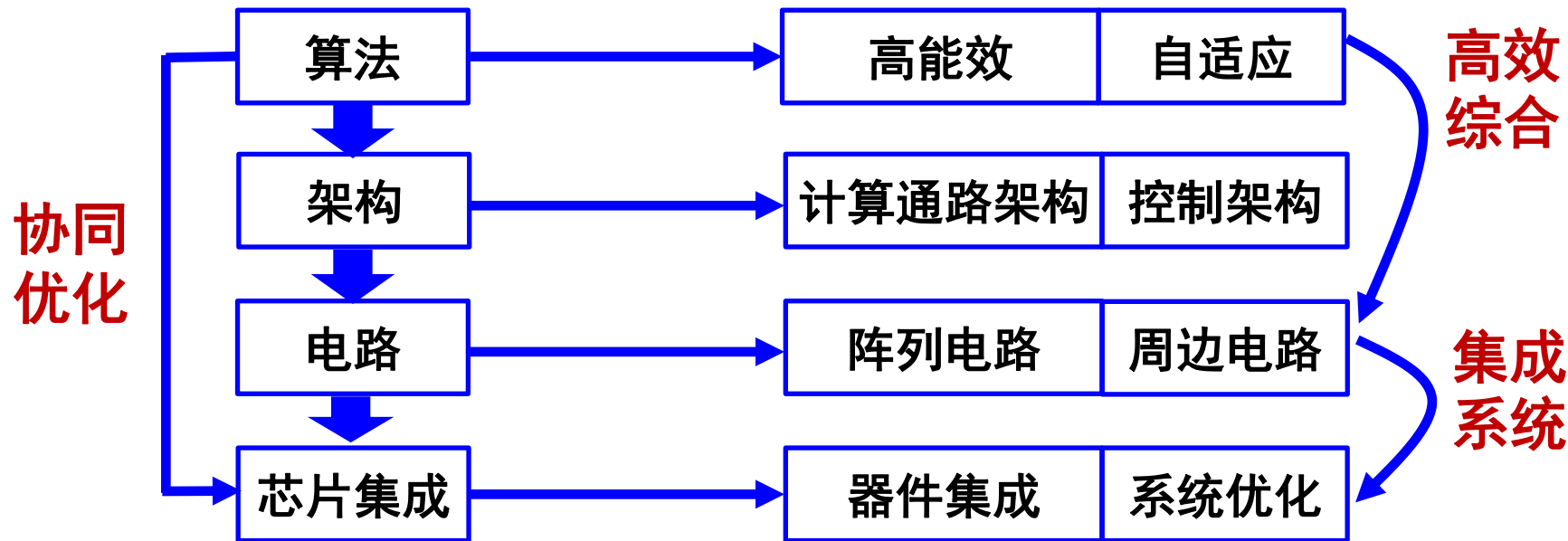
Task	Dataset	Model	Mode	Quality
Keyword Spotting	Google Speech Commands	DS-CNN	Single stream	90% (Top 1)
Visual Wake Words	Visual Wake Words Dataset	MobileNetV1 0.25x	Single stream	80% (Top 1)
Image classification	CIFAR10	ResNet-8	Single stream	85% (Top 1)
Anomaly Detection	ToyADMOS	Deep AutoEncoder	Single Stream	0.85 (AUC)

All MLPerf Tiny benchmarks are single stream, meaning they measure the latency of a single inference. The benchmarks also measure the model quality, which is either accuracy or AUC depending on the benchmark. MLPerf Tiny also enables optional energy benchmarking.

课程目标：算法-电路协同优化



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讲课时间和考试方式



讲课

周二下午1:30，有可能会有工业界报告延长一大节课

1-2个工业界报告

每周阅读一篇经典论文

考试方式

每5周一个Lab，共三个Lab，Project为Lab 3的提高要求

没有期中考试和期末考试

/ 授课计划 (Tentative)



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- 1) Introduction of DNN
- 2) DNN的原理
- 3) Quantization (Lab 1 out)
- 4) Pruning (剪枝)
- 5) Kernel Computation & Computing Flow
- 6) Lab 1 Presentation
- 7) Accelerator & Case Study (TPU&Ascend) (Lab 2 out)
- 8) Mapping of DNN & Toolchain (Layer Fusion)
- 9) Data Movement & Storage Hierarchy (Compression)
- 10) Lab 2 Presentation
- 11) 校庆+五一
- 12) Codesign Sparse-aware Arch & NAS (Lab 3 out)
- 13) Compute-in Memory (CIM)
- 14) End2End Deployment & ALP
- 15) Training
- 16) Conclusion



- One required paper to read every week.
- The review questions guide you through the paper reading process.
 - What are the **motivations** for this work?
 - What is the **proposed solution**?
 - What is the work's **evaluation** of the proposed solution?
 - What is your **analysis** of the identified problem, idea, and evaluation?
 - What are **future directions** for this research?
 - What **questions** are you left with?



Three Labs

Lab 1: 量化与剪枝实验

Lab 2: 矩阵计算阵列设计实验

Lab 3: 神经网络映射与稀疏推理实验

Project

Propose and validate some research ideas based on three Labs

/ 成绩说明



阅读论文：20%
5篇阅读报告

实验成绩：45%
Lab 1：20%； Lab 2：25%

Lab 3 + 大作业：35%
做一些研究探索

课程参与度：5%（加分）
课程提问、参与课题研究等



- If you turn in someone else's work as if it were your own, you are guilty of academic misconduct. This includes problem sets, answers on exams, lab exercise checks, project design, and any required course turn-in material.
- Also, if you knowingly aid in academic misconduct, you are guilty.
- We have software that compares your submitted work to others.
- However, it is okay to discuss with others lab exercises and the project (obviously, okay to work with project partner). Okay to discuss homework with others. But everyone must turn in their own work.
- If we catch your academic misconduct, you will get negative points on the assignment: It is better to not do the work than to cheat! If it is a midterm exam, final exam, or final project, you get an F in the class. All cases of academic misconducts reported to the office of student conduct.

GPGPU与深度学习算法的共同发展历史，未来10年？

ECE1742S: Programming using CUDA, 2008, U of T

Convolutional Neural Networks for Object Classification in CUDA

Alex Krizhevsky (kriz@cs.toronto.edu)

April 16, 2009

Convolutional Neural Networks for Object Classification

Alex Krizhevsky

Department of Computer Science, University of Toronto

akrizhevsky at gmail.com

Abstract I implemented a convolutional neural network with one layer of convolution. I tested it on the CIFAR-10 dataset, which consists of 6000 32×32 colour images in each of 10 classes. The convolutional net does well on the classification task and takes roughly 140x less time to train than a CPU implementation.

- Presentation:  (pdf).
- Report:  (pdf).
- Source code:  (zip).

1 Introduction

Here I will present my implementation of a simple convolutional neural network in CUDA. The network takes as input a 32×32 colour image and produces as its output one of ten possible class labels. The convolution operations, which account for 90% of the time required to train this network, are 125-150x faster on the GPU than on an Intel Core 2 Duo 2.4GHz.

9 Conclusion

It works! The next step is to build convolutional nets with multiple layers of convolution.

http://www.eecg.toronto.edu/~moshovos/CUDA08/arx/convnet_report.pdf



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Artificial Intelligence

“The science and engineering of creating intelligent machines”

John McCarthy, 1956

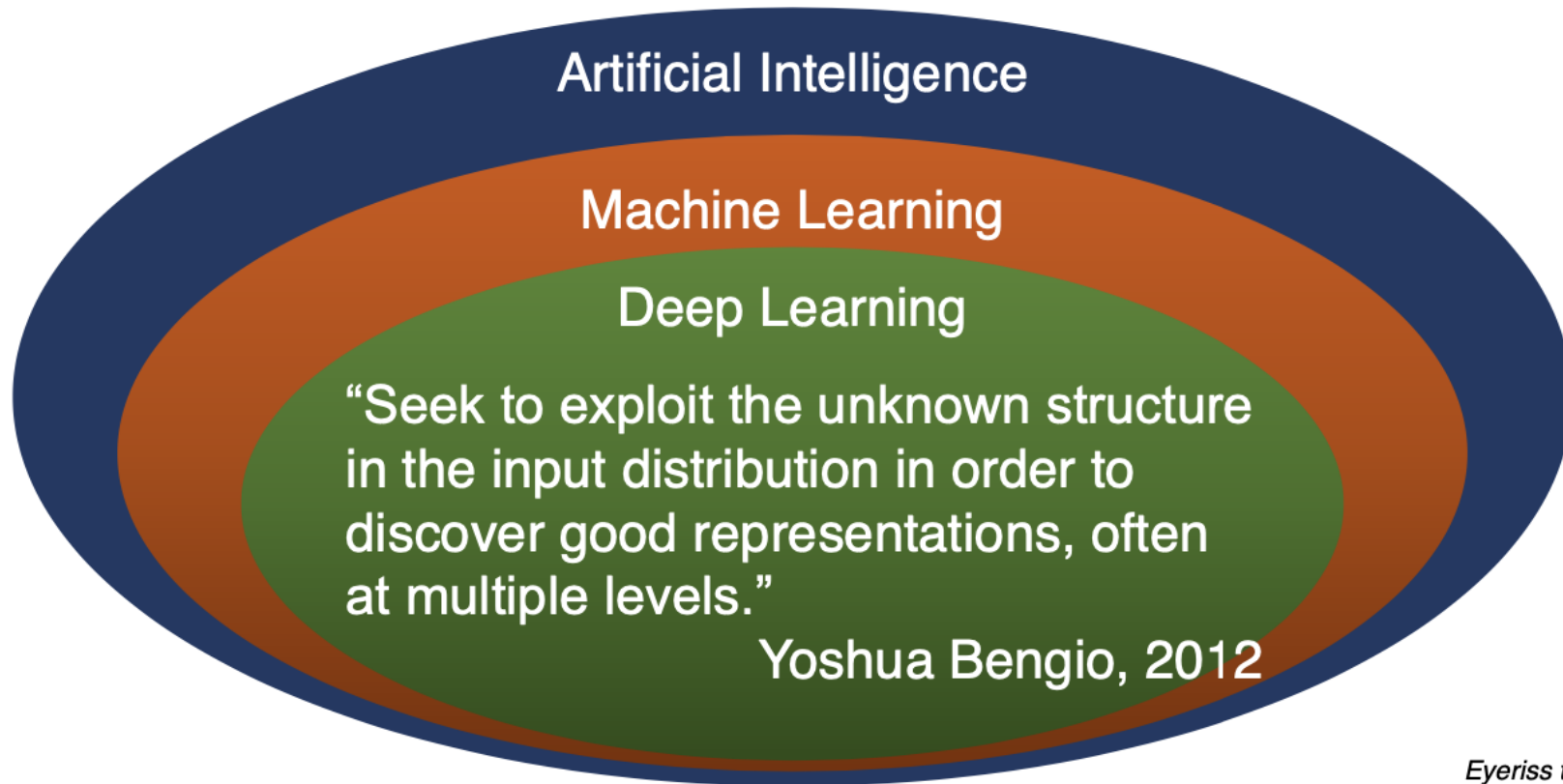


Artificial Intelligence

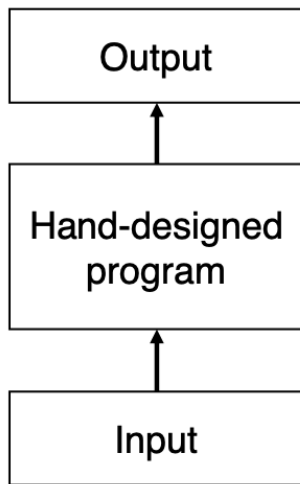
Machine Learning

“Field of study that gives computers the ability to learn without being explicitly programmed.”

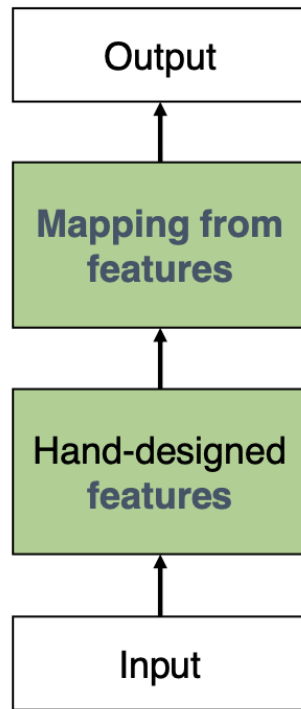
Arthur Samuel, 1959



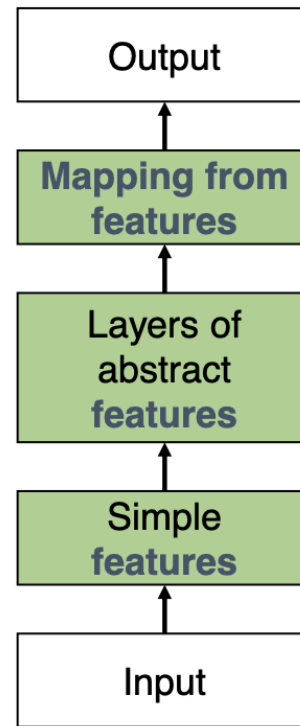
/ 各种类型的AI系统



**Rule-based
system**



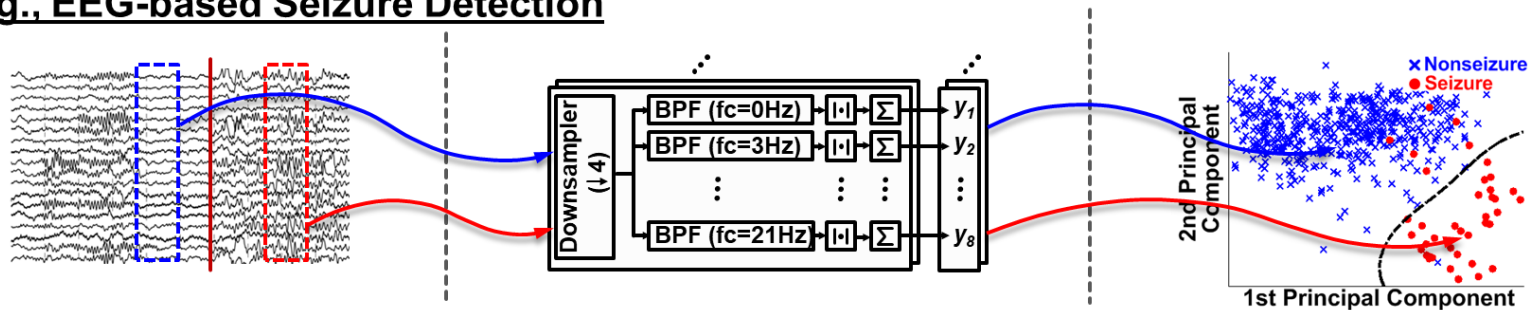
**Classic
machine
learning**



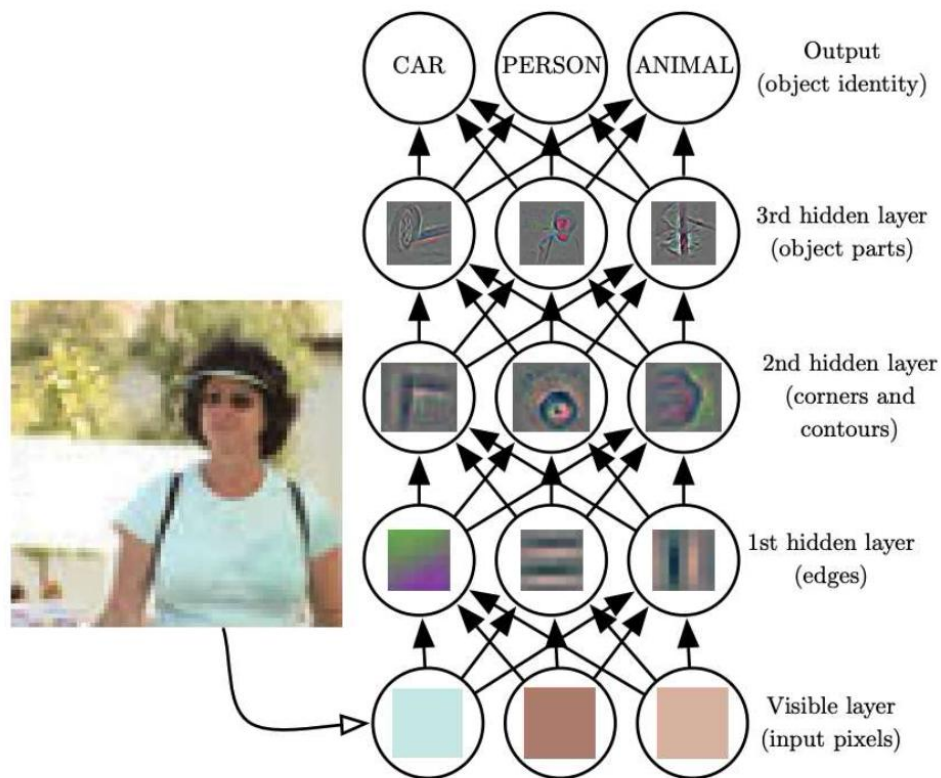
**Deep
learning**



E.g., EEG-based Seizure Detection



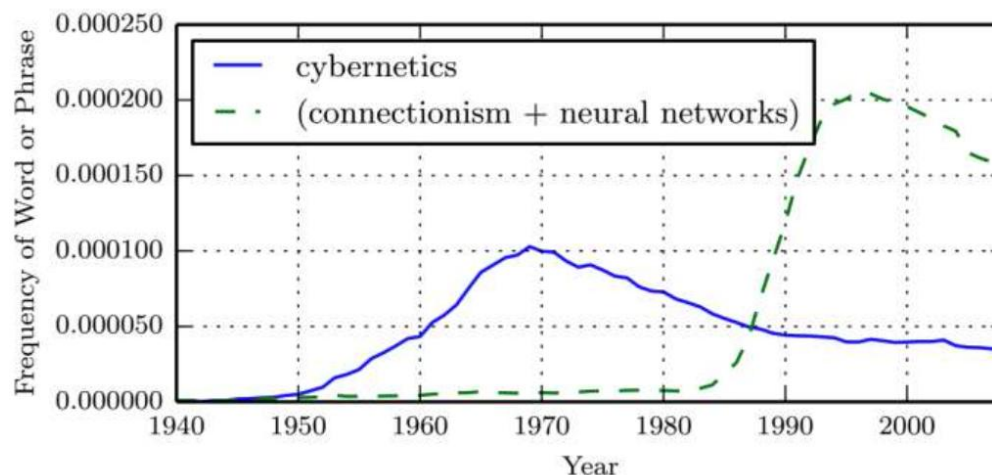
[H. Jia, VLSI'17]



深度学习不同命名阶段



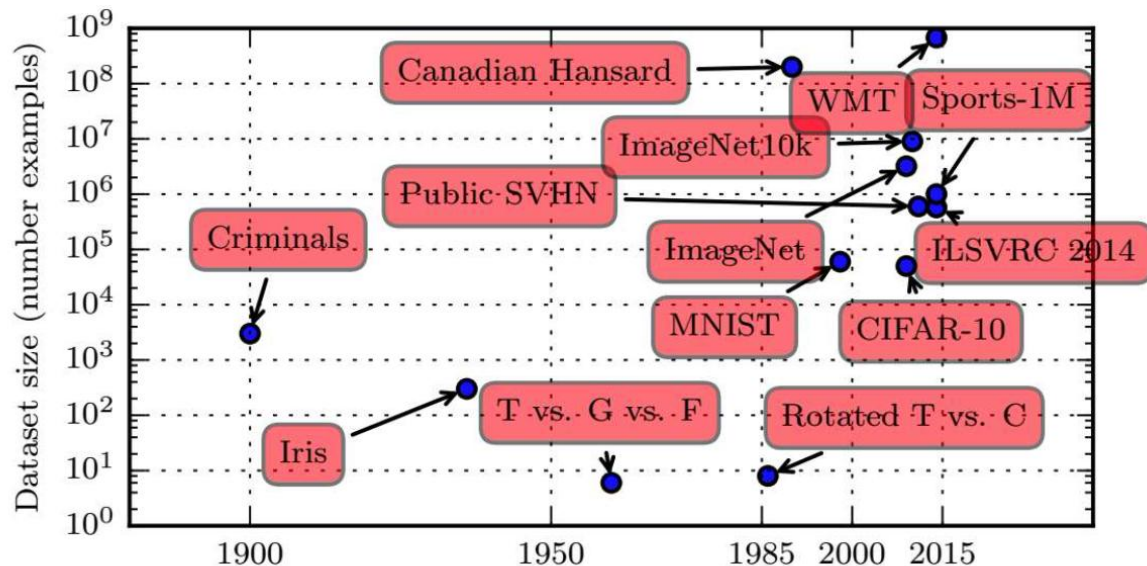
- Three waves of neural networks
 - Cybernetics
 - (1940s – 1960s)
 - Connectionism
 - (1980s – 1990s)
 - Deep learning
 - (2006s – now)
- Diminished role of neuroscience
 - Simply do not have enough information about the brain



不断增加的数据集合



- The size of datasets has expanded remarkably over time.
- The age of “Big Data” has made machine learning easier.



/ 不断增加的数据集合

- 60,000 32x32 images
 - 50,000 training
 - 10,000 test
- 10 classes
 - 6,000 images/class

airplane



automobile



bird



cat



deer



dog



frog



horse



ship



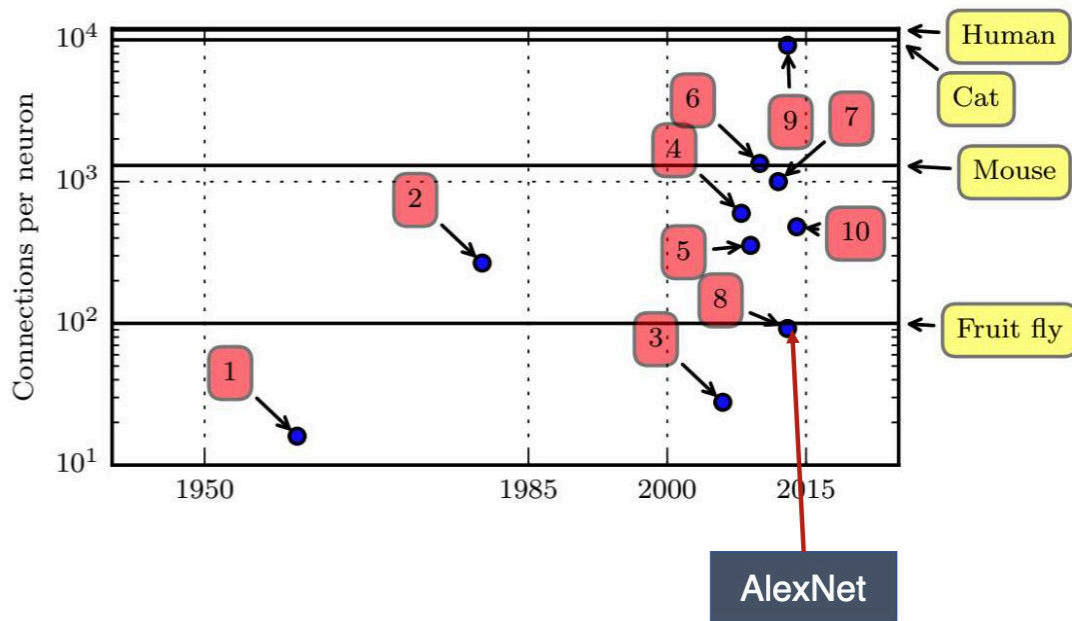
truck



不断上升的模型尺寸



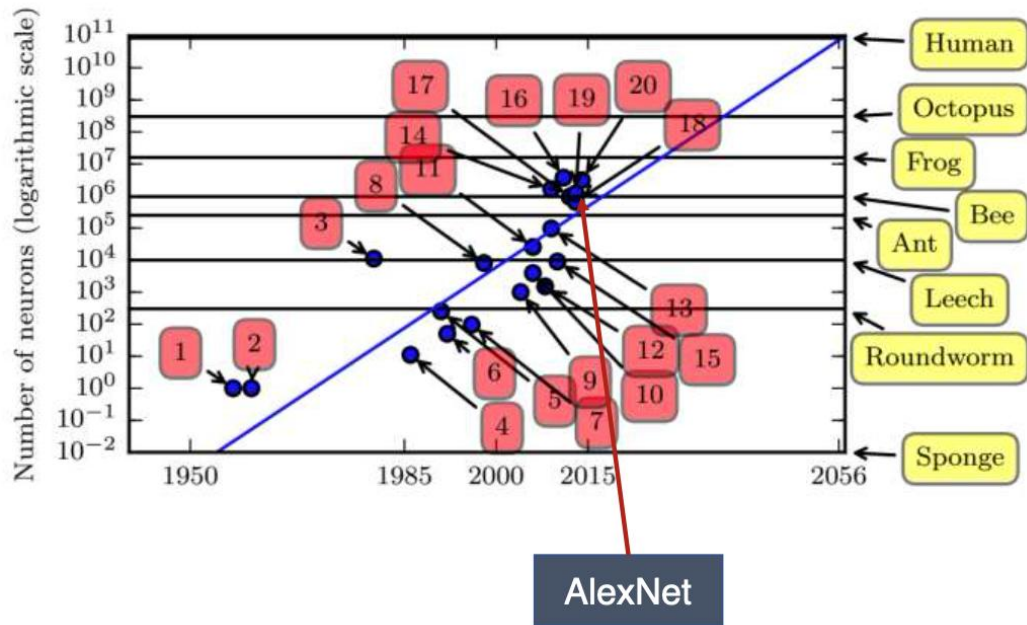
- Model size:
 - # of connections / neuron
 - # of neurons
- Largely due to the availability of faster hardware (CPUs and GPUs)
- Expect to continue



不断上升的模型尺寸



- Model size:
 - # of connections / neuron
 - # of neurons
- Largely due to the availability of faster hardware (CPUs and GPUs)
- Expect to continue





- “A computer program is said to ***learn*** from ***experience (E)*** with respect to some ***task (T)*** and some ***performance measure (P)***, if its performance on T, as measured by P, improves with experience E.”,
Tom Mitchell, 1998
- Example: spam classification
 - Task (T): Predict emails as spam or not spam.
 - Experience (E): Observe users label emails as spam or not.
 - Performance (P): # of emails that are correctly predicted.



- Suppose a tumor classification program watches which tumor is being marked as “benign” or “malignant” by doctors, and it learns how to better predict benign/malignant tumors.
- What is the task T in this setting?
 - A. The number (or fraction) of tumors correctly classified as benign/malignant
 - B. Watching doctors label tumors as benign or malignant
 - C. Classifying tumors as benign or malignant
 - D. Non of the above. This is not a machine learning problem.

/ 设计一个机器学习算法



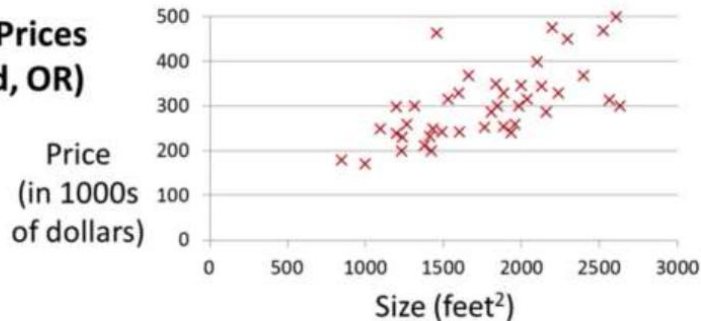
- Nearly all ML algorithms can be described as particular instances of a simple recipe:
 - A dataset $\rightarrow$ Experience (E)
 - A cost (loss) function $\rightarrow$ Performance Measure (P)
 - A model + An optimization method $\rightarrow$ Task (T)
- Use this recipe to see the different algorithms:
 - As part of a taxonomy of methods for doing related tasks that work for similar reasons
 - Rather than as a long list of algorithms that each have separate justifications

线性回归问题



- Dataset:
 - (x, y) where x is size and y is price
 - m training examples
- Cost function:
 - Mean Squared error
 - $MSE = \frac{1}{m} \sum_{i=0}^m (h(x_i) - y_i)^2$
- Model:
 - $h = w_0 + w_1 * x$
- Optimization method:
 - Solve for where its gradient is 0.
 - Gradient descent

Housing Prices
(Portland, OR)



Size in feet ² (x)	Price (\$) in 1000's (y)
2104	460
1416	232
1534	315
852	178
...	...

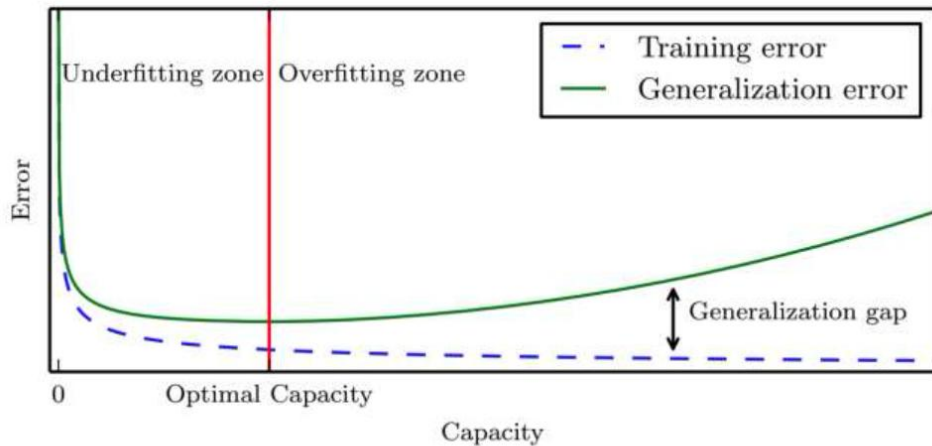


- ML tasks are usually described in terms of how the ML system should process an example.
- A dataset is a collection of many examples.
- ML algorithms can be broadly categorized as ***supervised*** and ***unsupervised*** by what kind of dataset they process.
 - Supervised: each example of the dataset is associated with a label or target
 - E.g., Classification, regression
 - Unsupervised: experience dataset without labels
 - E.g., Clustering
 - Reinforcement: Not a fixed dataset, interact with an environment

/ 数据集合



- Generalization:
 - Algorithm performs well on new, previously unseen inputs (not just those on which the model was trained).
- Partition the dataset into:
 - Training set: where the model was trained
 - Test set: where the trained model was tested
- Overfitting:
 - Large gap between training and test errors

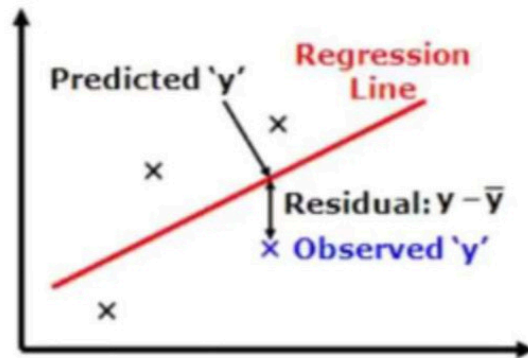


/ 成本函数



- Measures the performance of a ML model for given data
- Quantifies the error between predicted and expected values in the **training** set.

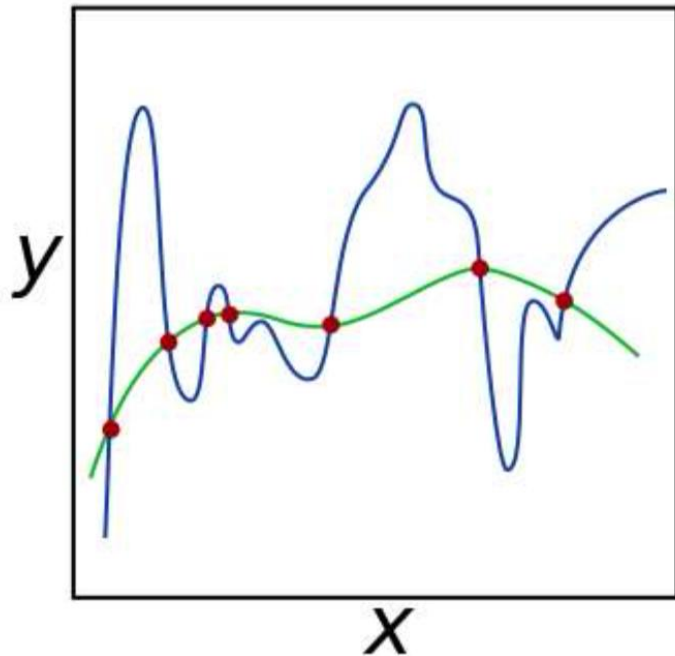
$$CF = MSE_{train} = \frac{1}{m} \sum_{i=0}^m (\hat{y}_i - y_i)^2$$



成本函数



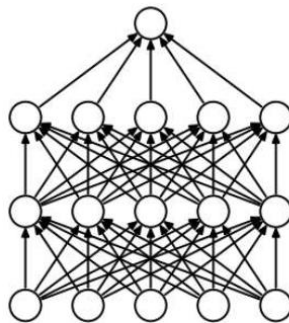
- Regularization: modifications to reduce the generalization error but not the training errors
 - Weight decay (L2/L1 regularization)
 - Expressing preferences of smaller weights
 - $CF = MSE_{train} + \lambda \sum_i w_i^2$ (L2)
 - $CF = MSE_{train} + \lambda \sum_i |w_i|$ (L1)



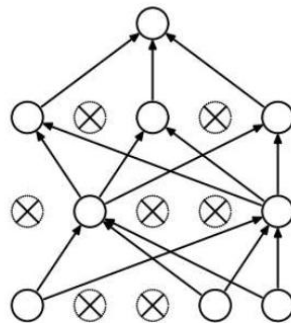
/ 成本函数



- Regularization: modifications to reduce the generalization error but not the training errors
 - Dropout
 - Temporally remove nodes from network
 - Train a large ensemble of models that share parameters



(a) Standard Neural Net



(b) After applying dropout.

/ 优化问题

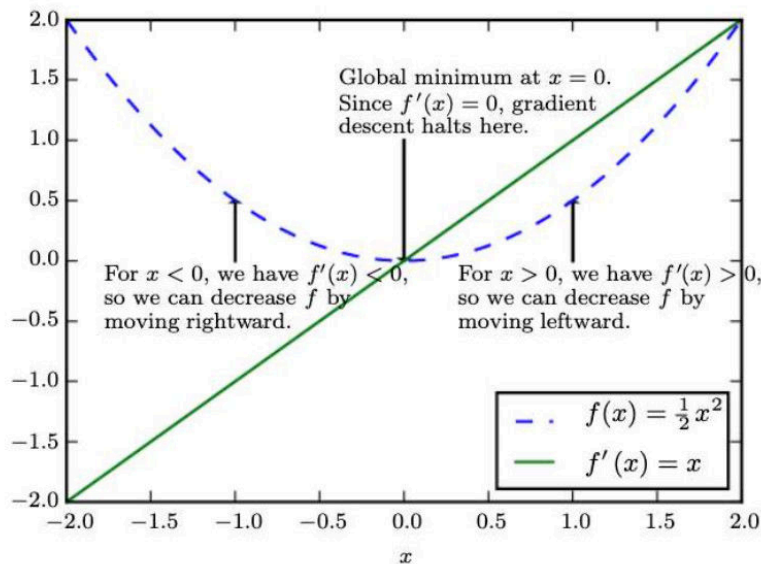
- Follow the slope



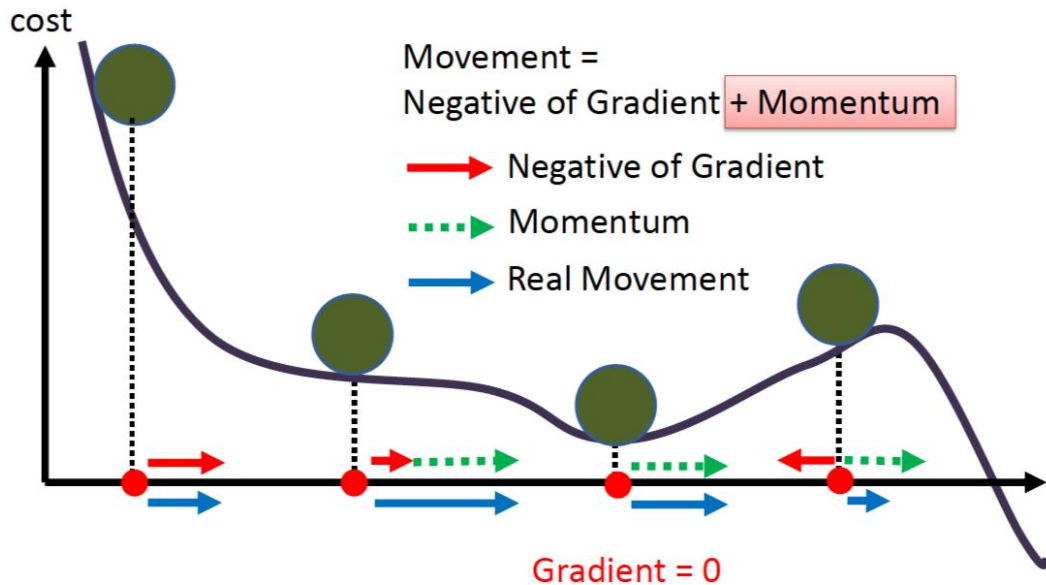
优化问题



- Method to minimize the **cost function** by updating weights
- Gradient descent:
 - Iteratively moving in the direction of steepest descent as defined by the negative of the gradient
- Stochastic gradient descent (SGD)
 - To handle large training sets
 - Only run a subset of the training sets (i.e., batch/minibatch) for each update
 - Easier to converge



- Momentum:
 - Prefers to go in a similar direction as before



SGD

$$x_{t+1} = x_t - \alpha \nabla f(x_t)$$

```
while True:  
    dx = compute_gradient(x)  
    x -= learning_rate * dx
```

SGD+Momentum

$$v_{t+1} = \rho v_t + \nabla f(x_t)$$

$$x_{t+1} = x_t - \alpha v_{t+1}$$

```
vx = 0  
while True:  
    dx = compute_gradient(x)  
    vx = rho * vx + dx  
    x -= learning_rate * vx
```

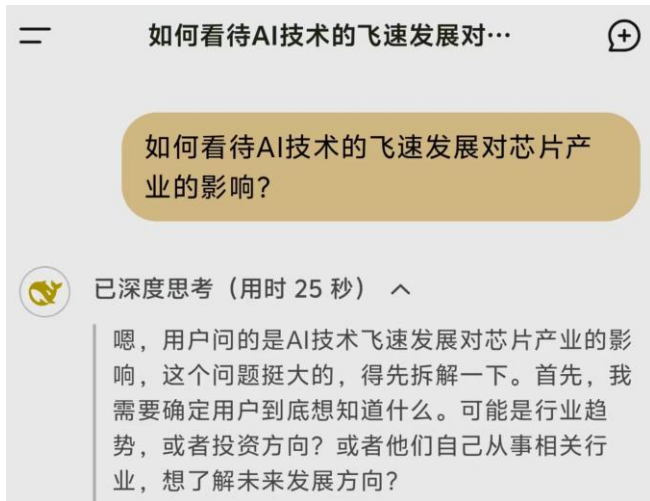
/ 深度学习的挑战



- The curse of dimensionality
 - Generalizing to new examples become exponentially more difficult when working with high-dimensional data
 - Challenging to learn complicated functions in high-dimensional spaces



deepseek-R1 (2025)



面对巨大的数据维度 (sequence length $\gg 10000$), 软硬件的协同优化变得尤其重要

OpenAI Sora (2024)



Prompt: A stylish woman walks down a Tokyo street filled with warm glowing neon and animated city signage. She wears a black leather jacket, a long red dress, and black boots, and carries a black purse. She wears sunglasses and red lipstick. She walks confidently and casually. The street is damp and reflective, creating a mirror effect of the colorful lights. Many pedestrians walk about.



- 1) Reading for this week
AlexNet Paper
Submit your review before next course.
- 2) Lab 1 will be posted after next course
Quantization of CNN



加州大学伯克利分校 EE290-2课程 Sophia Shao教授

寒武纪 智能计算系统 课程

三位助教帮助完成课件整理